

# Alejandro Sánchez Yalí

## AI Full Stack Engineer



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## SUMMARY

AI Full Stack Engineer with 8+ years designing and shipping web applications end to end and 3+ years integrating LLMs into production products. Sole architect and engineer of three live systems – an AI WhatsApp support platform, an AI-assisted recruitment platform and a surgical supply chain system – spanning Python/FastAPI backends, Next.js/TypeScript frontends, PostgreSQL and cloud deployment. M.Sc. in Mathematics and 11 years teaching, which shows up in design docs, code review and mentoring.

## TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, Effect, Three.js
- **Backend:** FastAPI, Django, Django REST Framework, Flask, NestJS, Node.js, Express, RESTful APIs, GraphQL, microservices
- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, PyTorch, TensorFlow
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, MySQL, Firebase, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Engineering:** Clean architecture, DDD, SOLID, TDD, Vitest, Jest, React Testing Library, code review

## EXPERIENCE

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| <b>AI Specialist</b><br><i>Lapzo</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Aug 2025 – Present<br><i>Remote, Mexico</i>      |
| <ul style="list-style-type: none"><li>– Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides</li><li>– Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time layer built with Next.js, NestJS and Firebase Realtime Database</li><li>– Lead member of the AI Committee, standardising AI development practices across the organisation</li></ul>                                                                                                                                                                                                                                         |                                                  |
| <b>Independent Software &amp; AI Consultant</b><br><i>Freelance</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Aug 2024 – Present<br><i>Remote</i>              |
| <ul style="list-style-type: none"><li>– Delivered end-to-end web applications for business clients across both halves of the stack: React.js/Next.js/TypeScript on the front, Python/FastAPI/Django and Node.js/NestJS behind it</li><li>– Lead engineer of Aluna (AI recruitment platform, see below) and builder of a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP with human escalation</li><li>– Led technical design sessions with clients and made the technology decisions for each project; code reviewer for a client's development team</li><li>– Deployed on AWS (EC2, S3, Docker) with CI/CD pipelines and Stripe payment integrations</li></ul> |                                                  |
| <b>Full Stack Engineer</b><br><i>Monadical</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Apr 2021 – Aug 2024<br><i>Remote, Canada/US</i>  |
| <ul style="list-style-type: none"><li>– Built and maintained full stack applications in Python (Django, FastAPI) and TypeScript (React.js, Next.js), including RESTful and GraphQL APIs</li><li>– Led frontend architecture, reusable component libraries and testing strategy (Vitest, Jest, TDD)</li><li>– Integrated AI/ML services and LLMs (OpenAI, LangChain, Whisper) into production web apps</li><li>– Led code reviews and mentored engineers on a distributed team across Canada, the US and Latin America</li></ul>                                                                                                                                                                                           |                                                  |
| <b>Full Stack Engineer</b><br><i>BCFort</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Aug 2018 – Oct 2020<br><i>Medellín, Colombia</i> |
| <ul style="list-style-type: none"><li>– Designed system architecture for blockchain platforms and built decentralised app frontends with React and Web3.js</li><li>– Developed analytics platforms with Python (Django) and Node.js, managing cloud infrastructure on AWS, GCP and IBM Cloud</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                  |

- Taught Machine Learning, Data Science and Mathematics with Python, NumPy, Pandas, TensorFlow and PyTorch
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research projects

## FEATURED PROJECTS

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- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, RAG, PostgreSQL, Redis, Next.js*
  - Sole architect and engineer of a multi-tenant SaaS where AI agents resolve routine customer inquiries in about 2 seconds and escalate the rest to human operators with full conversation context
  - Modular monolith with 10 DDD bounded contexts enforced by a lint rule in CI; multi-provider LLM layer (Groq default, OpenAI fallback) so switching providers is a config change
- **VitaStock – Surgical Supply Chain for Clinics**  – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
  - Sole architect and engineer of a platform for private surgical clinics in Colombia – 12 modules, 5 user roles, replacing spreadsheet-based inventory and procurement
  - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
  - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
  - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email

## OPEN SOURCE

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- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 486 GitHub stars across own open-source projects – top 4.9% of ranked GitHub profiles (GitRanks)

## TALKS & WRITING

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- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Building Plixiq: a Multi-Tenant WhatsApp AI Support Platform** – architecture case study (2026)

## EDUCATION

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**M.Sc. in Mathematics**  
*Universidad de Antioquia*

Aug 2010 - Sep 2013  
*Medellín, Colombia*

**B.Sc. in Teaching Mathematics and Physics**  
*Universidad de Antioquia*

Jan 2004 - Oct 2009  
*Medellín, Colombia*